Allenki Akshay

Software Engineer | AI & Full-Stack | Python, TypeScript, AWS

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Summary

Software engineer who builds full-stack and AI systems for production: a university website designed for 1M+ monthly requests, a speech recognition pipeline processing 10,000+ audio files a month, and event infrastructure for 30,000+ attendees. Currently a developer at Digital India Corporation (MeitY, Govt. of India) and building LearningPanda, an AI learning platform for schools, as a personal project. Most drawn to forward-deployed and AI engineering: working close to users, understanding the real problem, and shipping software that works for them.

Education

Vellore Institute of Technology - Andhra Pradesh — *Amaravati, IN*
Bachelors in Computer Science and Engineering — *2022 – 2026*

Narayana Junior College — *Hyderabad, IN*
M.P.C Telangana State Board of Intermediate Education — *2020 – 2022*

Krishnaveni High School — *Mancherial, IN*
Telangana State Board of Secondary Education — *2019 – 2020*

Experience

Junior Developer — *New Delhi, IN*
Digital India Corporation, MeitY, Govt. of India — *Sep 2026 – Present*

- Develop backend microservices in NestJS and TypeScript on AWS for Kisan Sarathi 2.0, a Government of India platform that connects farmers with agricultural advisories.
- Build and maintain APIs across the user, advisory, content and call services.
- Built activity logging across services using a message queue.
- Work on role management, online events and notification features, and write tests and API documentation.

Founding Engineer — *Remote, IN*
Oriflow — *Sep 2025 – Feb 2026*

- Built an agentic AI personal-growth platform that turns free-form user input into tasks, emotions and priorities, then guides the user along a personalised, gamified path that adapts as they progress.

Software Engineer Intern | *Live* — *Remote, IN*
Navigate Labs — *Sep 2024 – Feb 2025*

- Architected "Nexus Platform," an online code editor integrated with AI for educational institutions, making coding practice and learning more interactive.
- Built it with Next.js, Python and AWS services for a scalable architecture, and created an interactive dashboard for usage tracking.

Software Engineer Intern | *Certificate* | *Live* — *Amaravati, IN*
Vellore Institute of Technology, Andhra Pradesh — *Sep 2023 – Sep 2024*

- Led a team of five developers to build the official VIT-AP University website, with a system design capable of handling 1M+ monthly requests.
- Built a responsive front end with Next.js and Tailwind CSS and used Strapi headless CMS for content management, improving SEO by 40%.
- Deployed on AWS using S3, EC2, RDS, Amplify and Auto Scaling Groups, achieving 99.9% uptime with streamlined CI/CD pipelines.

Full-Stack Engineer Intern | *Certificate* — *Remote, IN*
Indian Institute of Technology, Bombay — *May 2023 – Apr 2024*

- Worked on OCR, automatic speech recognition (ASR) and transcript translation to make content accessible across multiple Indian languages.
- Built a React.js speech recognition interface, reducing customer wait times by 25% and improving first-call resolution.
- Led the integration of an audio transcription API that processed 10,000+ files monthly, cutting transcription times by 75%.
- Implemented a serverless Python Flask backend with Redis, accelerating deployment by 40% and improving response rates by 25% across the transcription and translation APIs.

Projects

LearningPanda – AI learning platform for schools | *Live* *May 2026 – Present*

- Architected an AI learning platform for schools as a monorepo: a FastAPI, PostgreSQL and Redis core API serving four Next.js portals (student, parent, school and super admin), with Celery workers and Docker deployments on AWS.
- Built a RAG tutor service with LangGraph and LangChain over a Milvus vector store, fed by a Celery ingestion pipeline that converts curriculum PDFs into searchable chunks with Docling and sentence-transformers.
- Reduced LLM cost and latency with response caching (24-hour TTL), request coalescing and request deduplication in the backend.
- Shipped "Board Teacher," an animated voice tutor that teaches on a virtual blackboard, with in-browser voice activity and turn detection using ONNX Runtime Web.
- Built LeetCode-style weekly coding labs where Python submissions are judged against hidden test cases in sandboxed AWS Lambda functions, with a CodeMirror editor and module-wise labs.
- Built an OCR answer-evaluation microservice on AWS SageMaker async endpoints, with a JWT validation API that checks a school's plan entitlement before evaluation.
- Added multi-version course videos with per-video and per-topic reviews, a student referral programme, a Jira-style task diary for students, upload security validation, and SEO, AEO and GEO improvements.

VITopia Website | *Code Repo* *Live*

- Built a QR-based ticket verification system for VITopia 2025 and VITopia 2024, streamlining entry for 30,000+ attendees over three days in 2025 and 20,000+ attendees over two days in 2024, and eliminating ticket fraud.
- Enabled seamless ticket purchases and automated bulk email distribution to 10,000+ students.
- Engineered a barcode and QR scanning system that reduced verification time by 80% and prevented duplicate entries, and optimised T-shirt distribution logistics with zero delays or mismatches.

AP CID Examination Portal | *Code Repo*

- Engineered a full-stack assessment platform for the AP Police Computer Instructor entrance exam, with secure Hall Ticket Number and date-of-birth login and a multi-stage test (typing speed test and MS Office task modules).
- Built a document submission system with Next.js and Tailwind CSS on the front end and a Python Flask backend handling concurrent Excel, Word and PPT uploads, storing data in PostgreSQL through Prisma ORM.
- Developed a post-exam reporting engine that gives candidates instant submission summaries while keeping evaluation scores isolated in the database for exam integrity.

Pothole Detection

- Created a real-time pothole detection system using **YOLOv8**, trained on 10,000+ images and deployed on **NVIDIA Jetson Nano** for edge inference.
- Improved road maintenance efficiency by enabling high-accuracy detection for proactive repairs.

Certificate Generator | *Code Repo* *Live*

- Launched a certificate generator web app for the CSI club with Next.js and Tailwind CSS, and personally resolved 15+ bugs in the first week to keep it stable on Vercel.

Resume Builder | *Code Repo* *Live*

- Designed and built a full-stack resume builder with Firebase authentication and Redux state management, reaching 50 sign-ups in its first week.

Technologies

Languages: Python, TypeScript, JavaScript, Java.
AI / ML: LangChain, LangGraph, RAG, Milvus, AWS Bedrock, SageMaker, Whisper, YOLOv8, ONNX Runtime, sentence-transformers, n8n.
Libraries: NumPy, Pandas, Matplotlib, NLTK, spaCy, TensorFlow, Pydub, Pyannote.
Frameworks: FastAPI, Flask, Django, NestJS, Next.js, React.js, Vue.js, Node.js, Express.js, Celery, Prisma ORM, TypeORM, Headless CMS (Strapi, Sanity).
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, SQS, IVS, SageMaker, Amplify, CloudFront), Azure, GCP, Docker, Kubernetes, GitHub Actions.
Databases: PostgreSQL, MySQL, MS SQL, SQLite, MongoDB, Redis, OpenSearch.
Real-time Systems: WebSockets, Socket.IO, WebRTC.
Relevant Coursework: Data Structures & Algorithms, Operating Systems, OOP, Design Patterns, Microservices Architecture, Database Management Systems, Cloud Computing, Natural Language Processing, Computer Networks.
Soft Skills: Leadership, Communication, Client relations, Problem-solving, Learning agility.

Awards and Achievements

Acknowledged Contributor – EMNLP 2024 | *Paper Link* *Nov 2024*
"Enhancing ASR Cross-Lingual Proper Noun Recognition Using Large Language Models"

AWS Academy Cloud Foundations | *Badge* *Oct 2024*
AWS Academy Cloud Architecting *Oct 2024*

Leadership and Volunteering

Core Member (Technical Team Lead) *Amaravati, IN*
VITopia, VIT-AP University *Jan 2025 – Mar 2025*
Core Member *Amaravati, IN*
VITopia, VIT-AP University *Jan 2024 – Feb 2024*
Tech Connect Volunteer *Mumbai, IN*
Techfest, IIT Bombay *Dec 2023*
Registration Coordinator *Amaravati, IN*
VITopia, VIT-AP University *Feb 2023*